

ShipmentSure: Predicting On-Time Delivery Using Supplier Data

1. Introduction

Our project, “ShipmentSure: Predicting On-Time Delivery Using Supplier Data”, aims to build a machine learning model that can predict whether a shipment will reach on time or not based on supplier and product-related factors. As the first step, we performed Exploratory Data Analysis (EDA) to understand the dataset, explore feature distributions, identify relationships between variables, and check for class imbalance in the target variable (Reached.on.Time_Y.N).

Dataset size: 10,999 records with 12 features (8 numerical, 4 categorical).

Target variable: Reached.on.Time_Y.N (0 = On time, 1 = Late).

2. Univariate Analysis

We examined individual features to understand their distributions:

i. Target Variable:

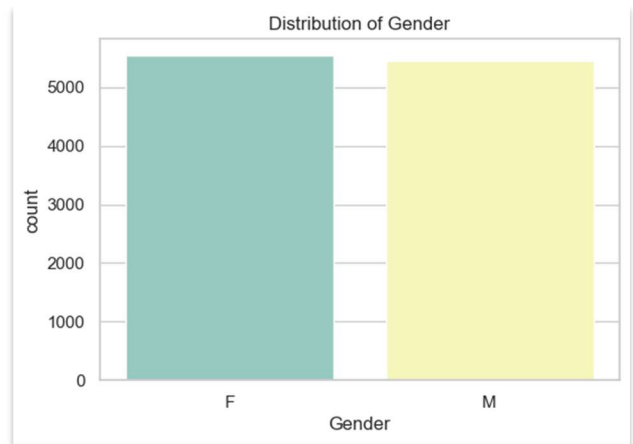
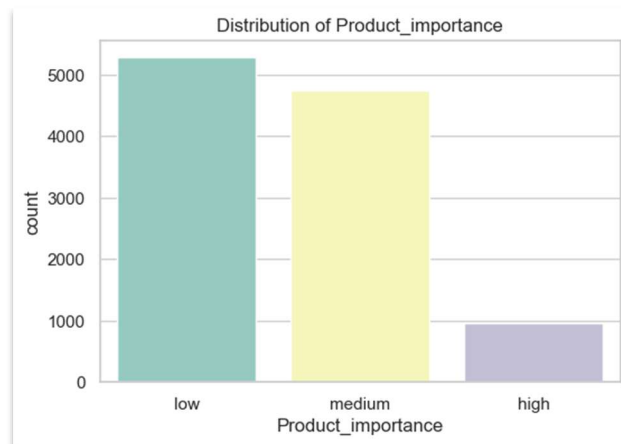
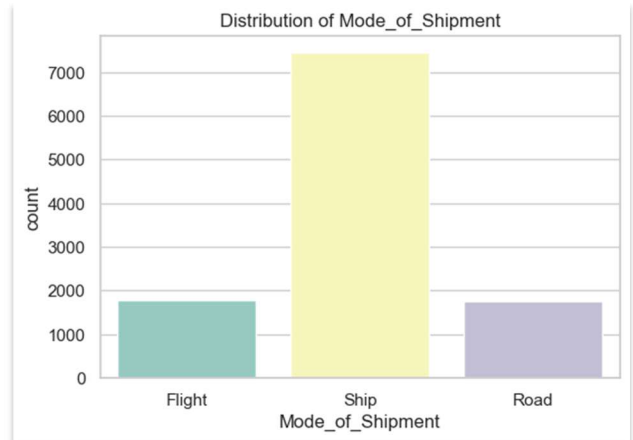
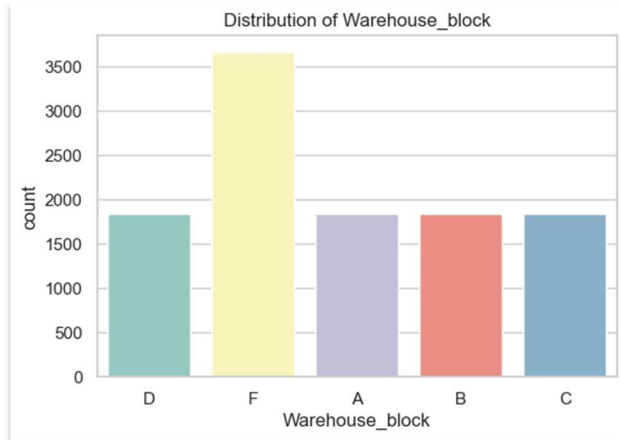
- Class 1 (Late): 6563 (59.7%)
- Class 0 (On Time): 4436 (40.3%)
- Shows mild imbalance toward delayed deliveries.



ii. Categorical Features:

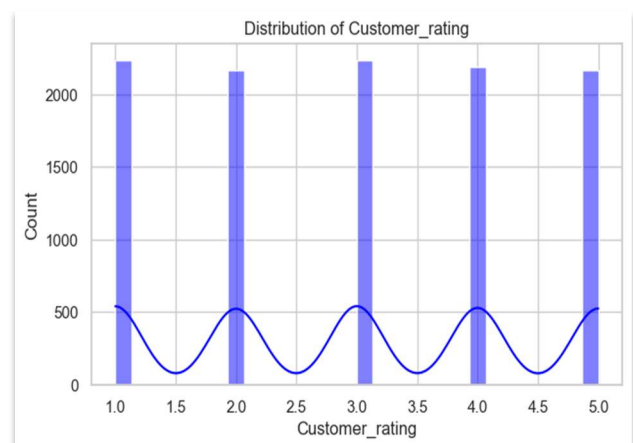
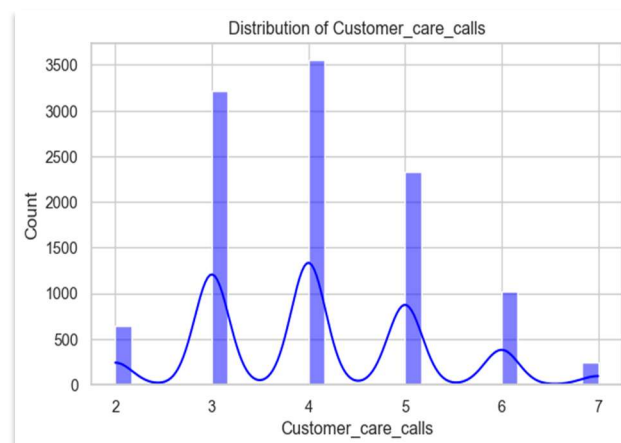
- Warehouse_block: Balanced distribution across blocks A-F.
- Mode_of_Shipment: Mostly “Ship”, followed by “Flight” and “Road”.
- Product_importance: Majority are “low” importance, fewer “medium” and very few “high”.

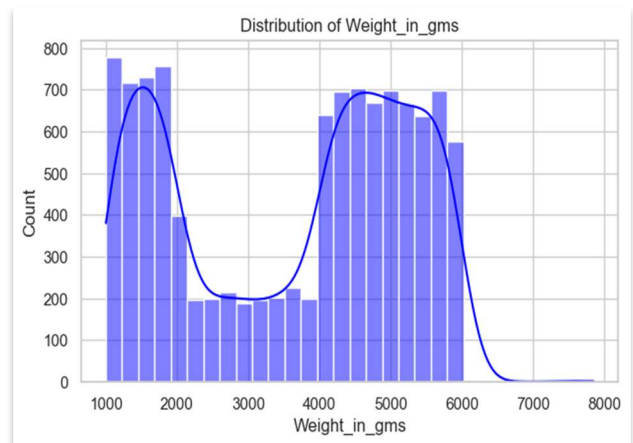
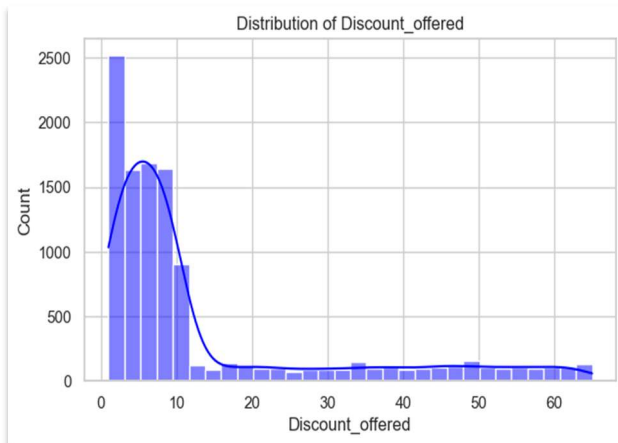
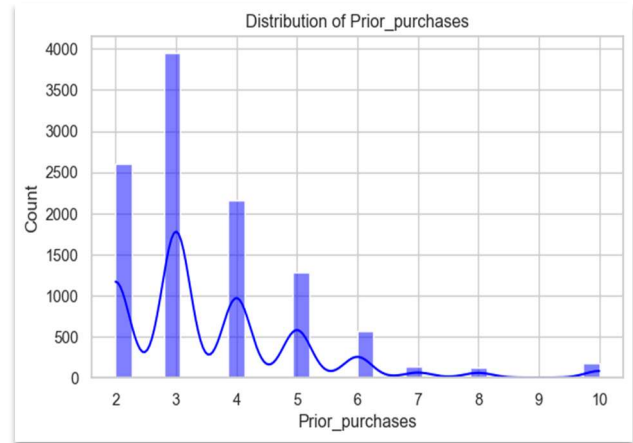
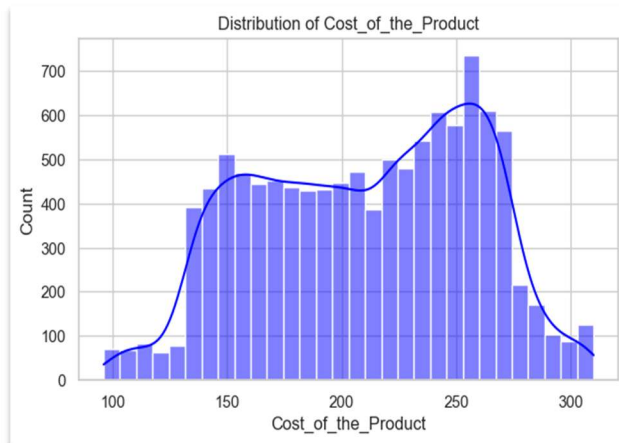
- Gender: Roughly balanced between Male and Female.



iii. Numerical Features:

- Customer_care_calls: Most values between 2–5 calls.
- Customer_rating: Fairly uniform from 1–5.
- Cost_of_the_Product: Wide spread with most values clustered in lower ranges.
- Prior_purchases: Majority have 2–5 past purchases.
- Discount_offered: Skewed toward lower discounts.
- Weight_in_gms: Normally distributed with a long tail.



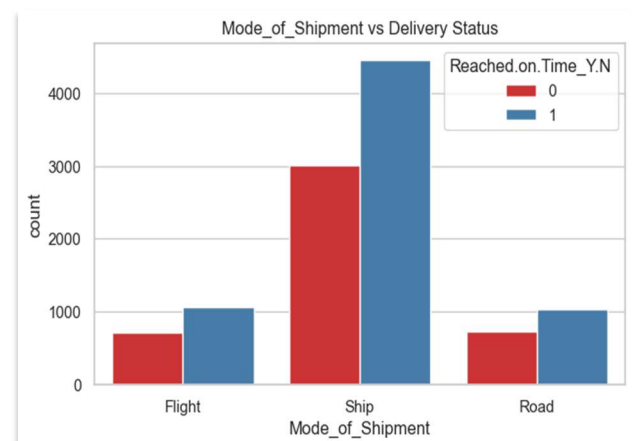
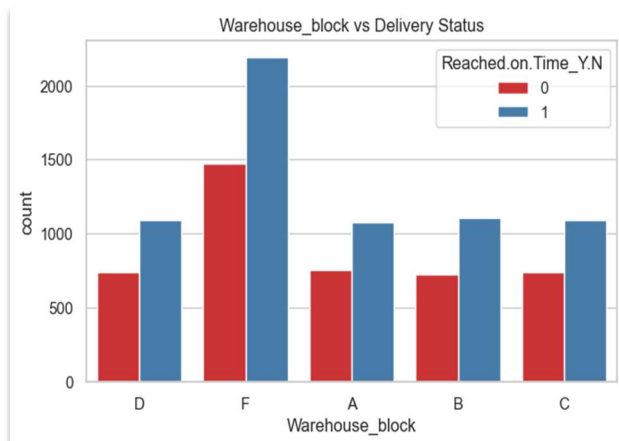


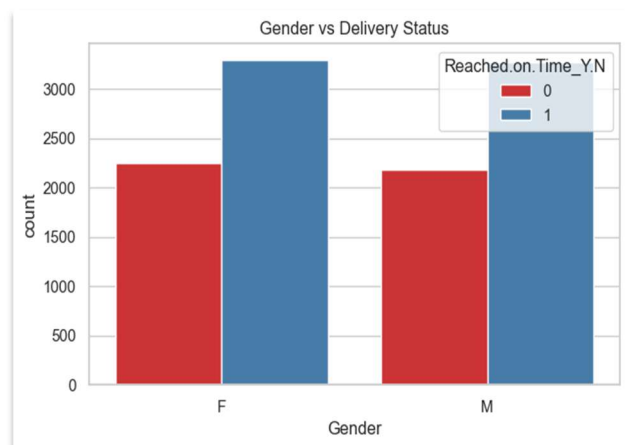
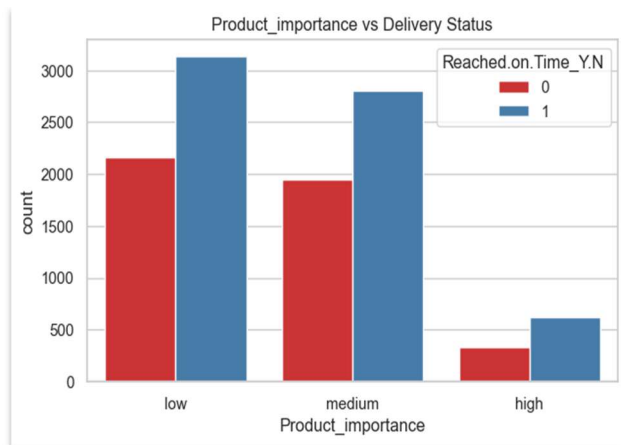
3. Bivariate Analysis

We explored how independent variables relate to the target variable:

i. Categorical vs Target:

- **Mode_of_Shipment:** Flights tend to have better on-time rates compared to shipping by road.
- **Product_importance:** High importance products tend to arrive on time more often.
- **Warehouse_block:** Slight variations, but not strongly predictive on their own.

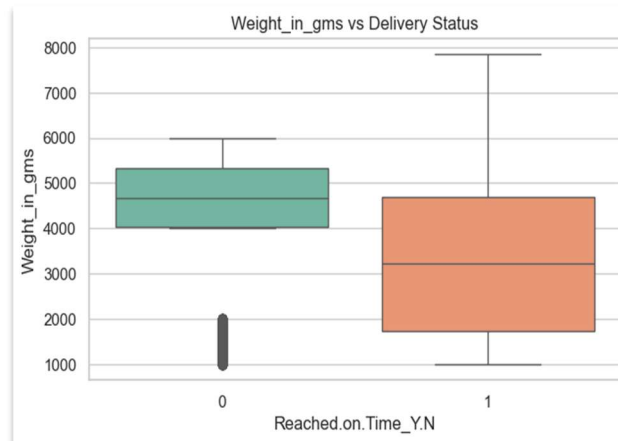




ii. Numerical vs Target:

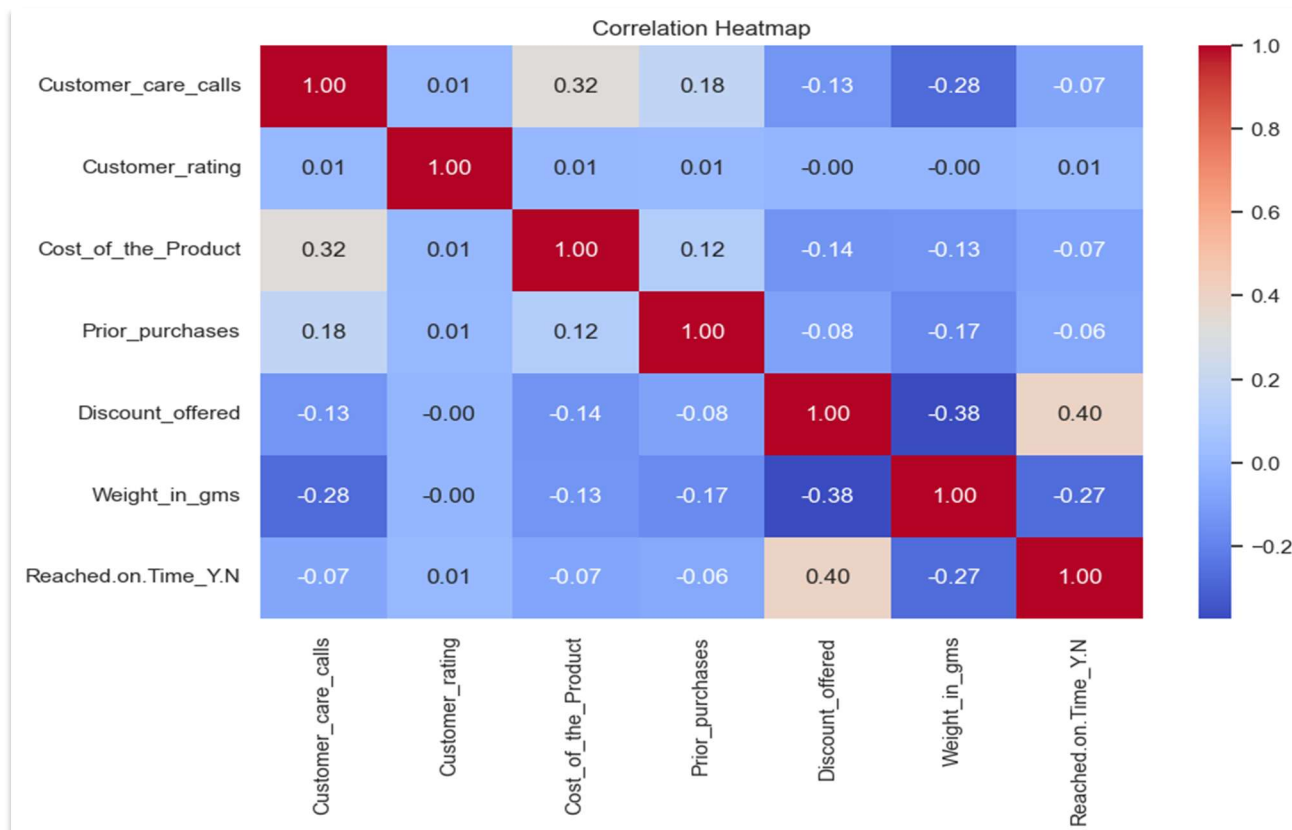
- Discount_offered: Higher discounts are associated with late deliveries.
- Customer_care_calls: Higher calls are linked to late deliveries.
- Weight_in_gms: Heavier products tend to have more delays.
- Customer_rating: Shows little variation with delivery status.





iii. Correlation Heatmap:

- Strong correlation between Weight_in_gms and Cost_of_the_Product.
- Negative relationship between Discount_offered and delivery status (late deliveries linked with higher discounts).



4. Class Imbalance Analysis

We analyzed whether the dataset is balanced for classification:

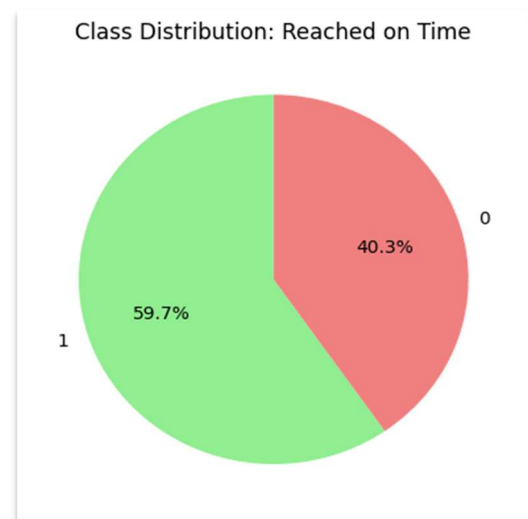
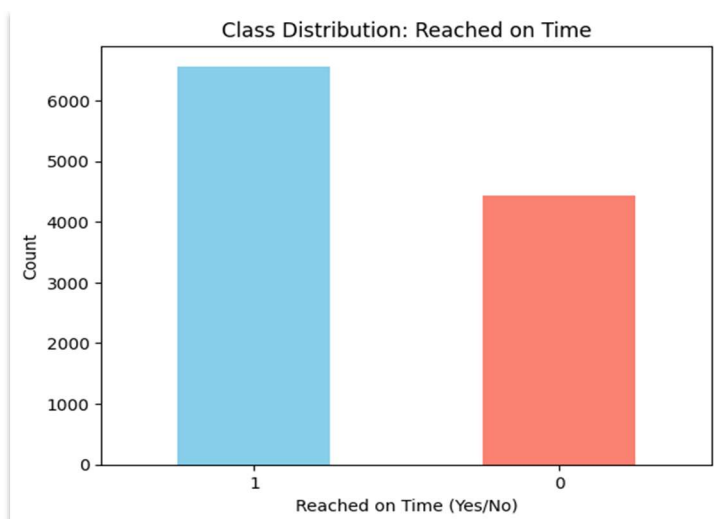
To examine class imbalance in the Reached.on.Time_Yes/No column, means we are checking whether the dataset has a roughly equal number of "Yes" and "No" values—or if one class dominates.

Distribution:

- Class 1 (Late): ~60%
- Class 0 (On Time): ~40%

Visualization:

- Bar plot and pie chart confirm imbalance.



- **Conclusion:**
Not heavily skewed, but imbalance-handling strategies (e.g., SMOTE, class weights) may be useful for modeling.

5. Work Completed So Far

- Data loading and cleaning.
- Univariate analysis of categorical and numerical features.
- Bivariate analysis between features and the target variable.
- Visualization of distributions, boxplots, and heatmaps.
- Class imbalance detection and analysis.